

Transfer Learning in Deep Learning: Using MobileNetV2 to Classify Flower Images with Less Training Data

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GitHub Link: <https://github.com/gurralalaxminarasimha/MachineLearning-Project>

1. Introduction

1.1 Context

Image classification represents a prevalent application of deep learning. It entails categorizing an image, for instance, determining whether a flower image depicts a daisy, dandelion, rose, sunflower, or tulip. Convolutional Neural Networks (CNNs) are extensively employed for this task due to their capability to learn visual patterns directly from picture pixels. Convolution and pooling procedures in CNNs progressively extract salient features, including edges, colors, textures, forms, and object structures.

Despite the efficacy of CNNs, training them from inception can be arduous. A model initialized with random weights must acquire all visual patterns solely from the provided training photos. This typically necessitates an extensive labeled dataset, adequate computer resources, and appropriate training duration. In instances of limited datasets, the model may lack sufficient variability in object appearance, illumination, backdrop, and image perspective. This may result in poor generalization on unfamiliar images.

1.2 Tutorial Focus

This tutorial emphasizes transfer learning using MobileNetV2. Transfer learning utilizes knowledge from a pre-trained model on an extensive image collection. The pretrained model offers a foundational feature extraction framework, eliminating the need to learn every visual characteristic from scratch. A novel classifier is subsequently incorporated for the target job.

The flower classification example illustrates this concept. A small CNN trained from inception is supplied for comparison, while MobileNetV2 serves as the transfer learning model. The focus is on comprehending the workflow: importing a pretrained network, immobilizing its foundational layers, incorporating a new classifier, and utilizing it for a novel picture classification challenge.

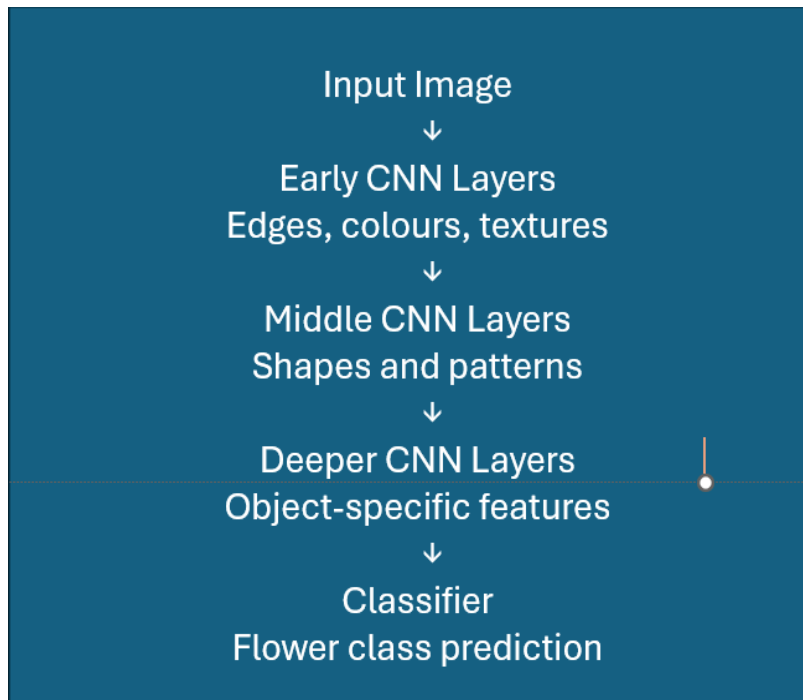
2. Theoretical Foundation of Transfer Learning

2.1 Feature Acquisition in Convolutional Neural Networks

Convolutional Neural Networks acquire picture features hierarchically. Initial layers typically identify fundamental visual attributes such as lines, color contrasts, and edges. The intermediate layers amalgamate these elements into more complex

patterns, including textures and forms. Deeper layers denote more task-specific characteristics, including object components and class-associated structures. The hierarchical structure renders CNNs adept for picture categorization, since the model transitions from basic pixel-level information to significant visual notions.

Figure 1: The CNN feature learning process, progressing from basic edges and colors to more complex, object-specific features utilized for flower classification.



A flow diagram illustrating the processing of a picture by a CNN through initial, intermediate, and deeper layers, culminating in a classifier that predicts the flower class.

2.2 Constraints of Training from Inception

A CNN initialized from the beginning possesses no pre-existing knowledge of images. All convolutional filters are initialized randomly, requiring the model to learn both low-level and high-level features only from the training dataset. This approach is excellent with extensive datasets, but it is less dependable with a limited number of photos.

Limited image databases may lack sufficient samples of diverse flower orientations, backdrops, lighting conditions, and visual changes. Consequently, a model developed from the ground up may exhibit delayed learning or may be prone to overfitting. Overfitting transpires when a model excels on training photos then falters with novel images, since it has committed individual instances to memory instead of acquiring generalizable patterns.

2.3 Principle of Transfer Learning

Transfer learning is predicated on the notion that knowledge acquired from one task might facilitate performance in another related task. In computer vision, numerous initial visual properties are universal. Edges, color transitions, corners, and textures are valuable across numerous image collections. A CNN trained on an extensive image dataset can consequently yield valuable feature representations for a smaller, novel job.

In transfer learning, the original classifier of the pretrained model is eliminated as it was tailored for the first dataset. The convolutional foundation is preserved due to its valuable feature extraction layers. A new classifier has been introduced for the additional classes. This method diminishes the necessity to acquire knowledge of each aspect from the outset, hence enhancing training efficiency.

3. MobileNetV2 as a Feature Extractor

3.1 MobileNetV2 Architecture

MobileNetV2 is an efficient convolutional neural network architecture tailored for computer vision applications where speed and model size are critical. It employs inverted residual blocks and linear bottlenecks to minimize computational expense while preserving efficient feature representation (Sandler et al., 2018). This renders it appropriate for a concise practical tutorial, as it may illustrate transfer learning without necessitating an extensive model.

3.2 Elimination of the Original Classifier

MobileNetV2 is typically trained for extensive picture classification tasks. The original top layer forecasts the categories from the initial challenge, rendering it unsuitable for flower categorization. Consequently, the model is loaded devoid of its upper classification layer. The residual convolutional base analyzes an image and generates feature representations. The features are subsequently input into a freshly developed classifier tailored for the five floral categories.

3.3 Immobilizing the Pretrained Foundation

Freezing constitutes a crucial phase in transfer learning. When the MobileNetV2 base is immobilized, its pretrained weights remain unchanged during the initial training phase. This preserves the valuable visual knowledge already embedded in the model. Only the recently incorporated categorization layer undergoes training. The frozen MobileNetV2 base functions as a feature extractor, while the new classifier learns to associate those features with the categories of daisy, dandelion, rose, sunflower, and tulip.

4. Dataset and Image Preparation

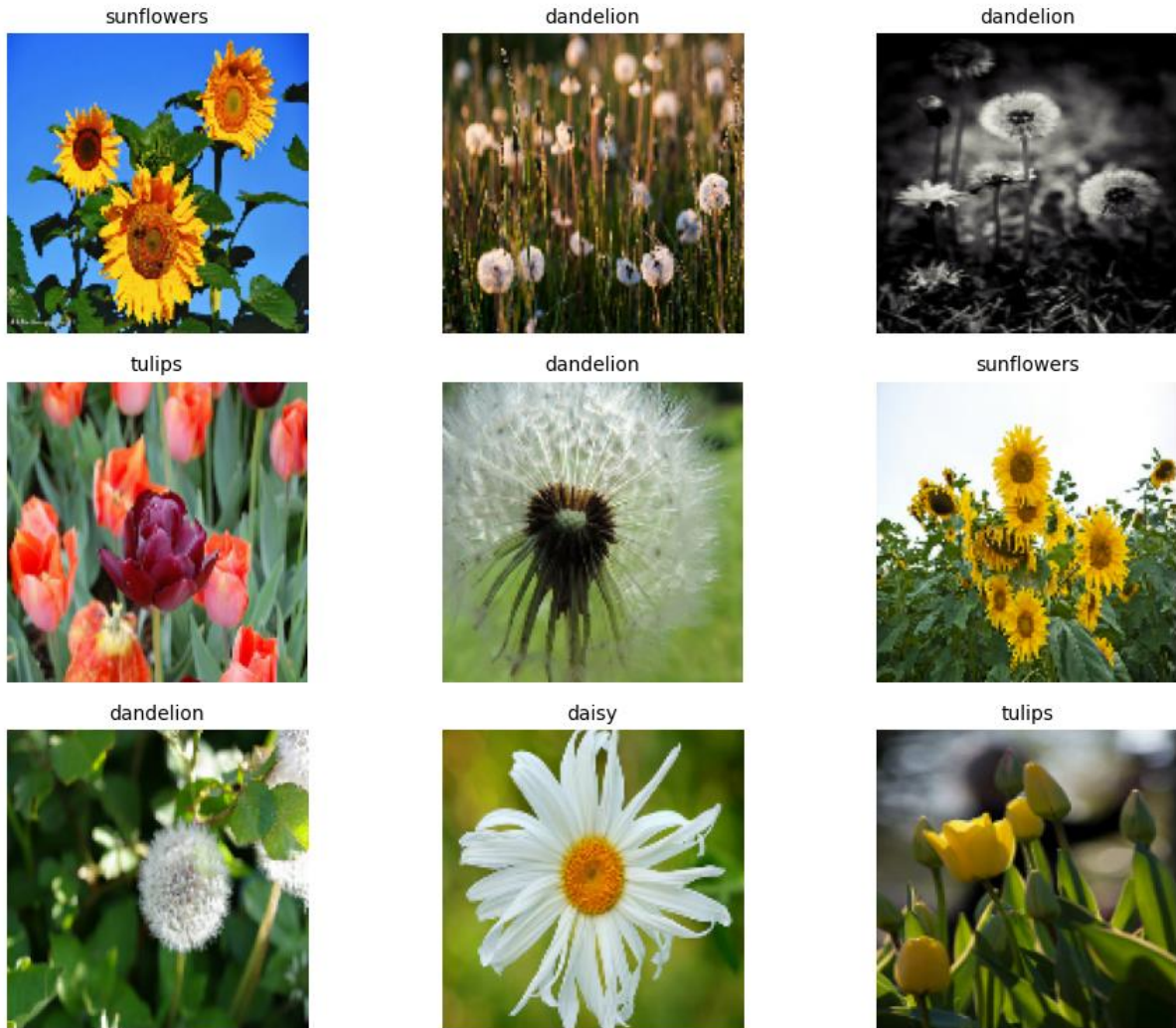
4.1 Floral Taxonomy Dataset

The lesson employs a floral image collection comprising five categories: daisies, dandelions, roses, sunflowers, and tulips. Each image is reduced to 128×128 pixels to ensure efficient notebook performance. The objective is multi-class image classification, wherein a single image is input into the model to predict a certain flower type.

The dataset is appropriate for illustrating transfer learning, as the classes exhibit distinct visual patterns, though they are not entirely straightforward. Flowers may differ in hue, form, dimension, backdrop, and perspective. Certain flowers may exhibit similarities, rendering the challenge beneficial for demonstrating how pretrained visual elements can enhance classification.

Figure 2: Illustrative flower photos from the dataset, demonstrating visual distinctions among daisies, dandelions, roses, sunflowers, and tulips.

Example Flower Images



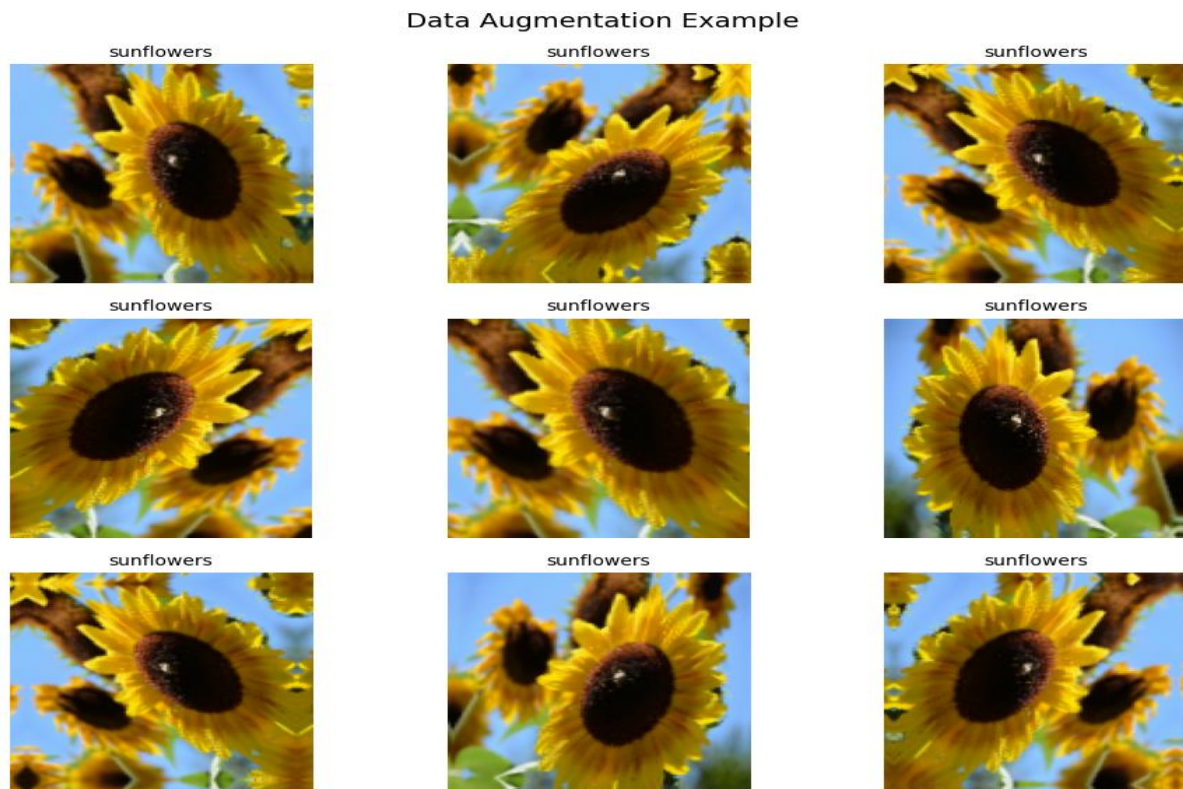
A grid of floral photos depicting examples from five categories: daisies, dandelions, roses, sunflowers, and tulips.

4.2 Data Augmentation

Data augmentation generates altered iterations of training images by the application of minor modifications. Random flipping, rotation, and zooming are employed in the notebook. These modifications do not alter the class label. A sunflower retains its identity regardless of minor rotations or zooms in its depiction.

Augmentation facilitates the model's acquisition of broader visual patterns. The model analyzes diverse iterations of analogous photos rather than relying on the precise position or orientation of a bloom in the training image. This can diminish memorization and facilitate improved generalization.

Figure 3: An illustration of data augmentation featuring rotated, magnified, and flipped variations of the identical floral image.



A grid displaying multiple augmented iterations of the identical floral image, encompassing rotated, zoomed, and flipped variations.

5. Convolutional Neural Network Developed from the Ground Up

5.1 Reference Model

A minimal CNN developed from the ground up serves as a baseline for comparison. This model initiates with arbitrary weights and learns exclusively from the flower dataset. It comprises convolutional layers, pooling layers, a global average pooling layer, and a final classifier. The convolutional layers extract visual patterns, whilst the classifier generates the final flower forecast.

The foundational model is deliberately uncomplicated. Its purpose is not to attain the highest possible accuracy, but to demonstrate a model's behavior in the absence of pretrained knowledge. The model has a more challenging learning challenge than MobileNetV2, since it assimilates all features from the provided flower photos.

5.2 Acquiring Knowledge from the Ground Up

At the commencement of training, the baseline CNN possesses no pre-existing knowledge of edges, forms, or floral structures. These must be acquired progressively from the training data. The model's learning is constrained due to a limited subset of images and a brief training duration. This substantiates the primary

comparison in the tutorial: training from inception necessitates that the model autonomously cultivates visual features, whereas transfer learning commences with a pre-existing, beneficial visual representation.

6. MobileNetV2 Transfer Learning Procedure

6.1 Importing the Pretrained Model

MobileNetV2 is initialized with pretrained weights and excludes its original classification layer. The convolutional base is retained due to its encapsulation of learned visual characteristics. Prior to the image being processed by MobileNetV2, the appropriate preprocessing function is employed to ensure the image format aligns with the expected input specifications of the pretrained model.

6.2 Incorporating the New Classifier

Subsequent to the extraction of visual features by the frozen MobileNetV2 base, the output is transmitted through a global average pooling layer and a dropout layer. The concluding dense layer forecasts one of the five floral categories. The new classifier constitutes the trainable component of the model in the initial transfer learning phase.

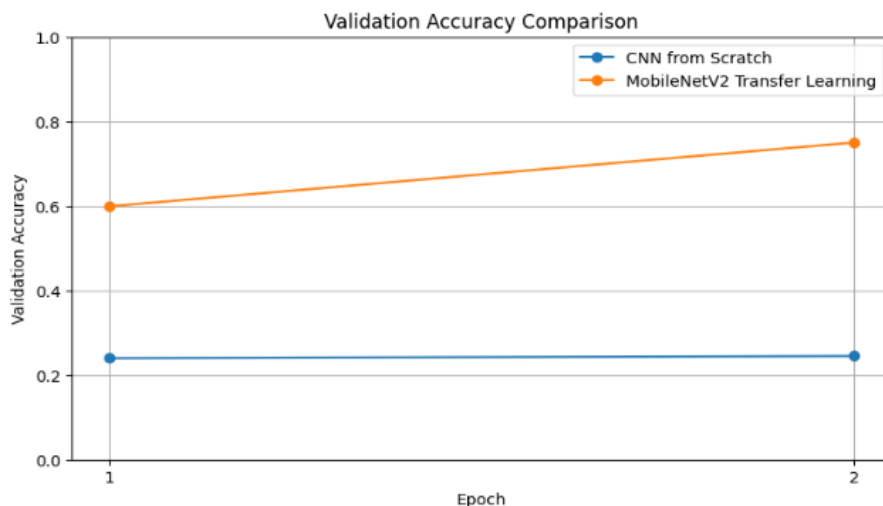
This process is referred to as feature extraction. The pretrained base offers general visual features, whereas the new classifier tailors those features for the flower classification task. This is a prevalent and pragmatic transfer learning strategy when the target dataset is inferior in size to the dataset employed for training the original model.

7. Learning Conduct and Exhibition

7.1 Brief Training Exhibition

The notebook employs a limited subset of the flower dataset and a reduced number of epochs to ensure rapid execution of the code. In this presentation, the tiny CNN, trained from inception, achieved a validation accuracy of 28.65% after two epochs, but the MobileNetV2 transfer learning model attained a validation accuracy of 67.19%. These figures are not disclosed as definitive production outcomes. They serve to demonstrate the disparity in learning behavior.

Figure 4: Comparison of validation accuracy between the CNN trained from inception and the MobileNetV2 transfer learning model.



A line chart illustrating the validation accuracy across two epochs for a CNN developed from inception and a MobileNetV2 transfer learning model. MobileNetV2 has superior validation accuracy.

Section 7.2: Interpretation

The disparity arises due to the two models commencing from distinct knowledge bases. The CNN initiates with arbitrary weights and must acquire all visual characteristics from the flower photos. MobileNetV2 encompasses valuable image representations acquired from a significantly larger image dataset. Consequently, the novel flower classifier may acquire knowledge more effectively due to enhanced feature representations derived from the pretrained basis.

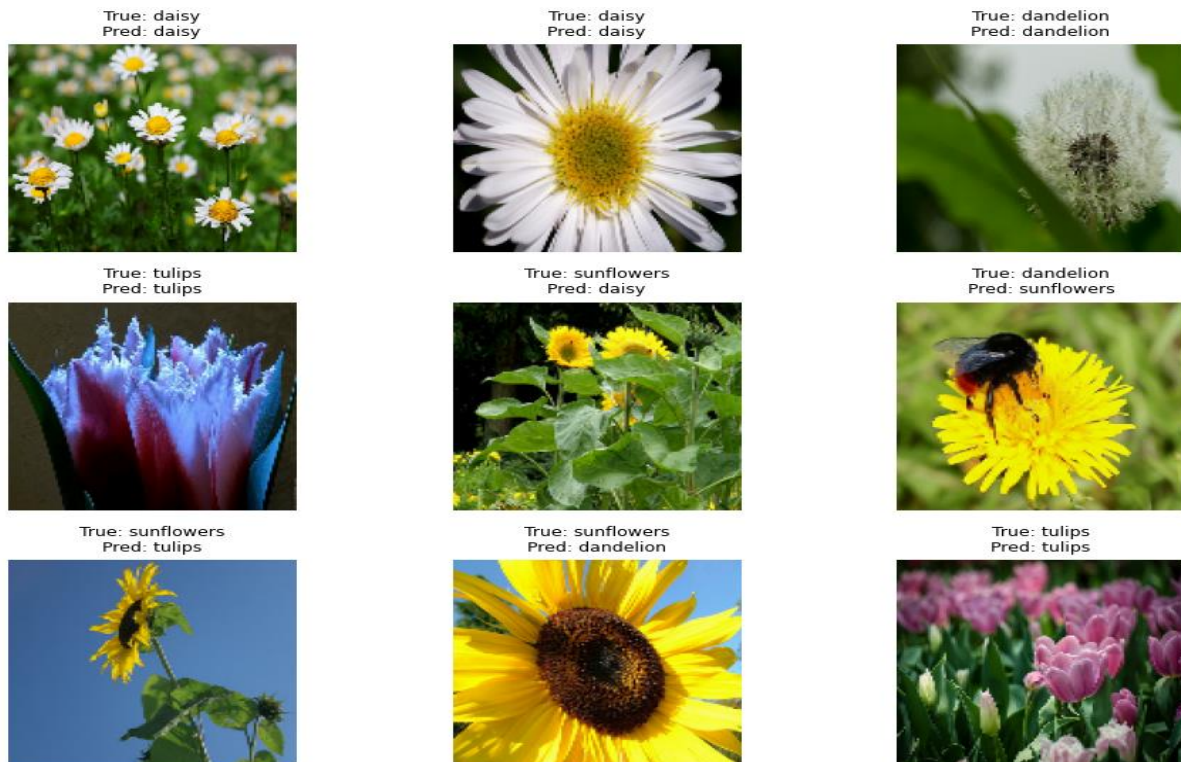
8. Exemplars of Prediction

8.1 Model Output

Examples of predictions illustrate how the trained MobileNetV2 model allocates labels to validation images. Each image is presented alongside the actual label and expected label. Accurate predictions indicate that the model has assimilated valuable visual indicators, including petal morphology, hue, and floral architecture.

Figure 5: Illustrative predictions from the MobileNetV2 transfer learning model, displaying actual and expected flower labels.

Example Predictions from Transfer Learning Model



Alt text: A grid displaying flower photos, each accompanied by its actual flower label and the anticipated label generated by the MobileNetV2 transfer learning model.

8.2 Constraints

Erroneous predictions are significant since they demonstrate the imperfection of transfer learning. Certain flower classes may exhibit analogous visual patterns, particularly when the image is ambiguous, clipped, or influenced by backdrop variability. Transfer learning can enhance learning efficiency; however, it does not eliminate the necessity for meticulous validation, appropriate data, and responsible interpretation.

9. Pragmatic Guidance

9.1 Suggested Procedure

A realistic transfer learning approach often commences with the selection of an appropriate pretrained convolutional neural network for the task at hand. The original classification layer is eliminated, and a new classifier is incorporated for the target classes. The pretrained base should typically be initially frozen, permitting just the new classifier to undergo training. If additional enhancement is required, fine-tuning may be pursued subsequently by unfreezing some top layers with a minimal learning rate.

9.2 General Considerations

The input photos must be appropriately scaled, and the preprocessing function for the selected pretrained model should be utilized. Data augmentation must be realistic and should not alter the image to such an extent that the class meaning is compromised. Model performance must be evaluated using both quantitative metrics and visual prediction instances, particularly when classes exhibit visual similarities.

10. Conclusion

Transfer learning is a potent method for picture categorization in scenarios when datasets are restricted or training time is constrained. This tutorial utilized MobileNetV2 as a static feature extractor, enabling the model to leverage pretrained visual features rather than learning all patterns from the beginning. The concise CNN comparison illustrated the utility of pretrained representations, as MobileNetV2 commenced with pre-existing knowledge of edges, textures, and forms. The tutorial illustrates that transfer learning can enhance the practicality of deep learning, contingent upon the meticulous use of preprocessing, augmentation, and prediction validation.

References

- Chollet, F. (2021) *Deep Learning with Python*. 2nd edn. Shelter Island: Manning.
- Deng, J. et al. (2009) 'ImageNet: A large-scale hierarchical image database', *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255.
- Pan, S.J. and Yang, Q. (2010) 'A survey on transfer learning', *IEEE Transactions on Knowledge and Data Engineering*, 22(10), pp. 1345–1359.
- Sandler, M. et al. (2018) 'MobileNetV2: Inverted residuals and linear bottlenecks', *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4510–4520.
- Yosinski, J. et al. (2014) 'How transferable are features in deep neural networks?', *Advances in Neural Information Processing Systems*, 27.